



Advanced Digital Image Processing Lecture #10

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Contents

- ❖ Pattern Recognition Approaches
 - ❖ Template Matching
 - ❖ PCA Method
 - ❖ Subspace Method
 - ❖ Knowledge-Based Approach
 - ❖ Neural Network
 - ❖ Hidden Markov Model (HMM)



Assignment: (Group1)

Presentation on Following Keywords

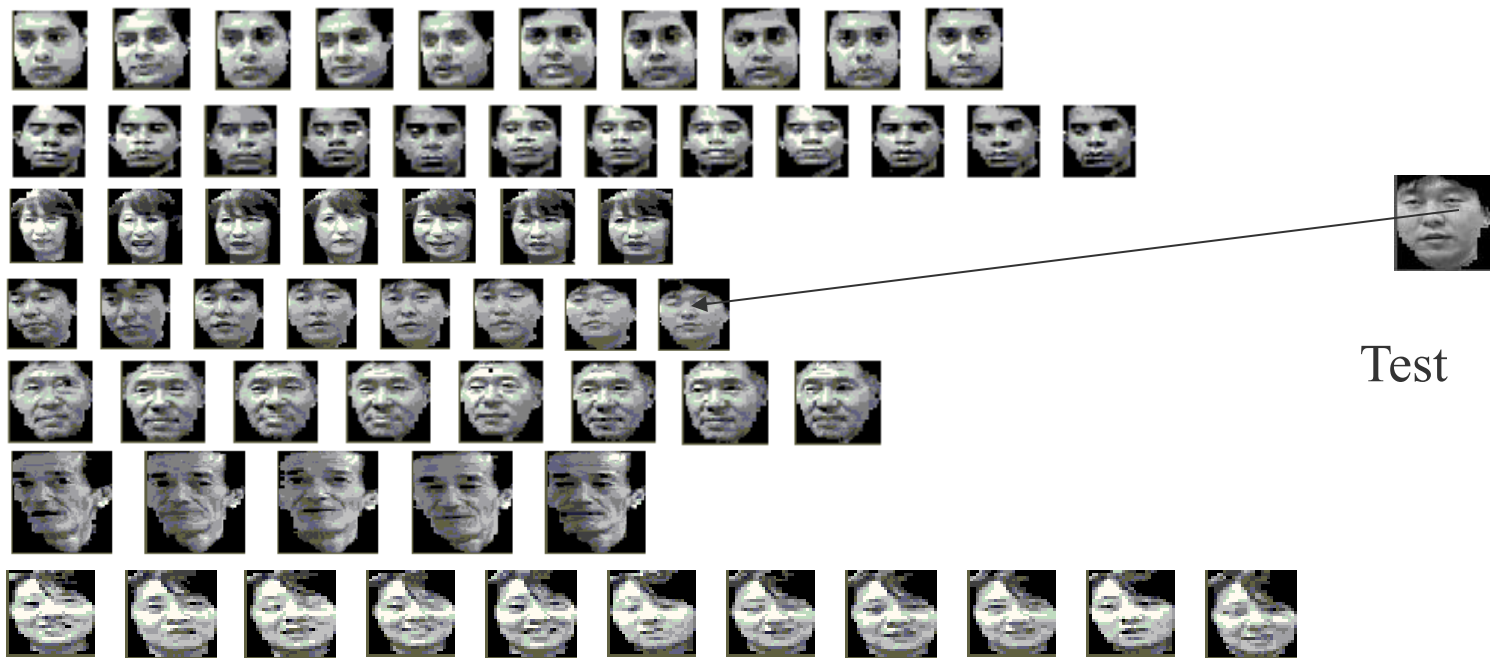
- ❖ Explain following terms using examples:
 - ❖ Mean, median
 - ❖ Variance
 - ❖ Covariance
 - ❖ Eigenvectors
 - ❖ Eigenvalues
 - ❖ PCA (principal component)

Principal Component Analysis (1 of 10)

- ❖ Prepare training images for face or predefined hand poses, $X_i(N \times N)$
- ❖ Calculate eigenvectors (u_m) from training images
- ❖ Select k-number of eigenvectors (u_k) corresponding to the highest eigenvalues to form PCA.

Principal Component Analysis (2 of 10)

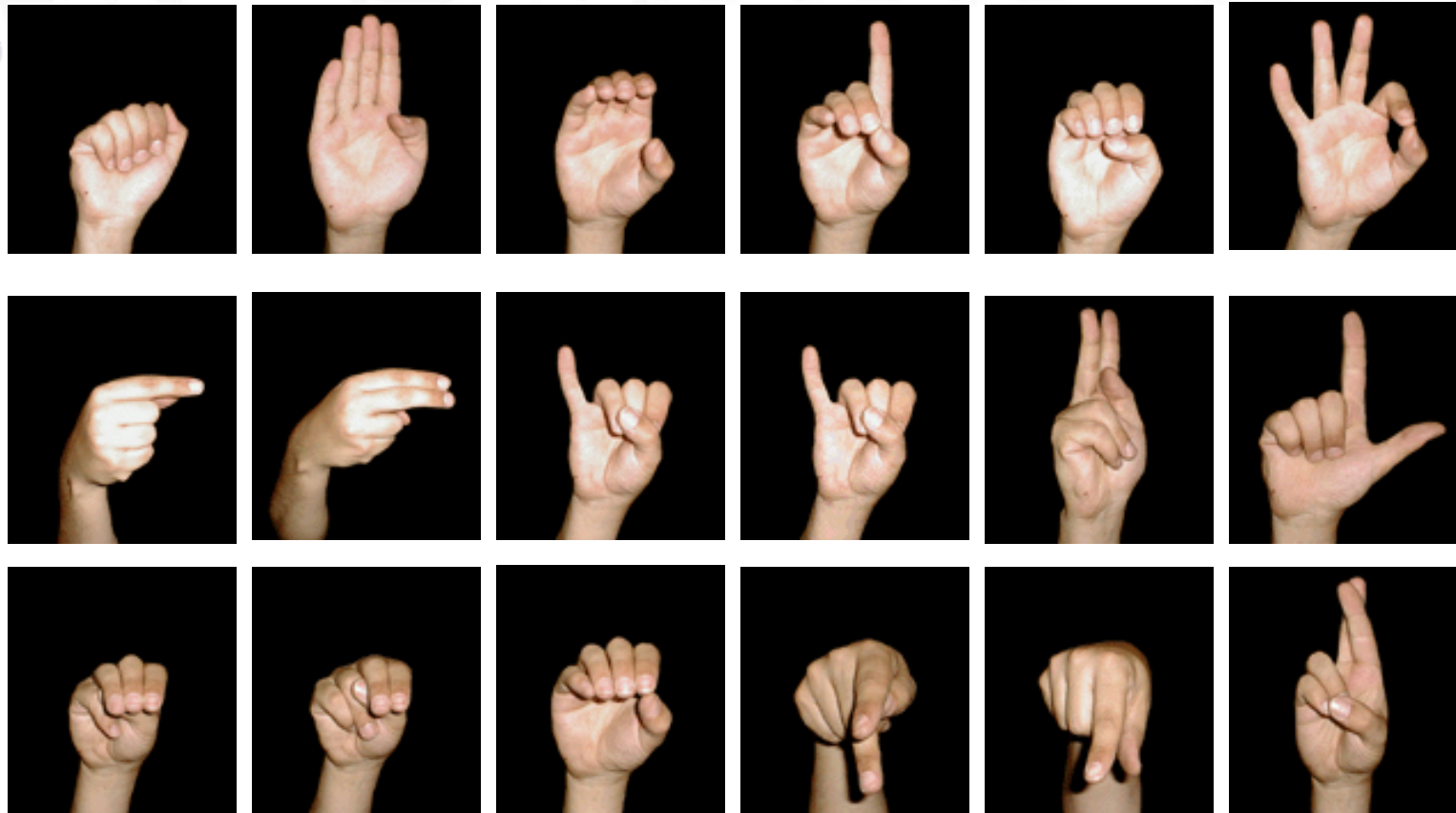
Example: Face Recognition



Training Images

Principal Component Analysis (3 of 10)

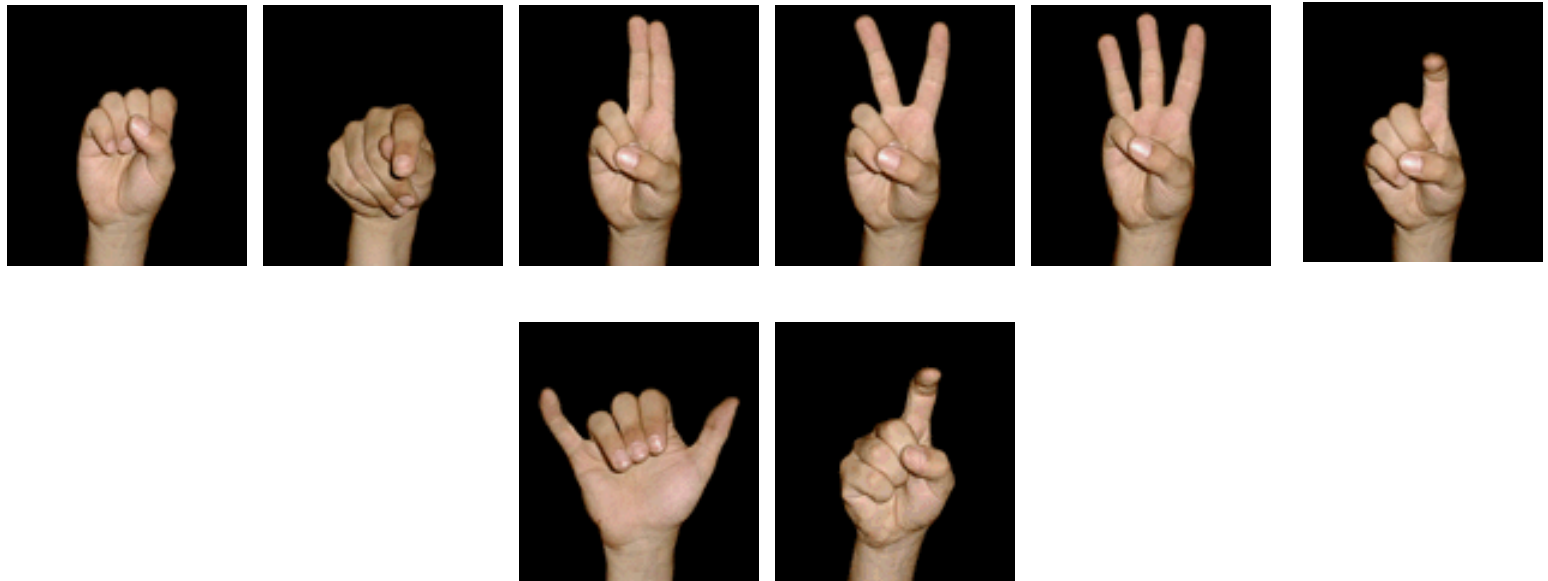
Example: ASL Characters Recognition



26 ASL Characters (A-R)

Principal Component Analysis (4 of 10)

Example: ASL Characters Recognition



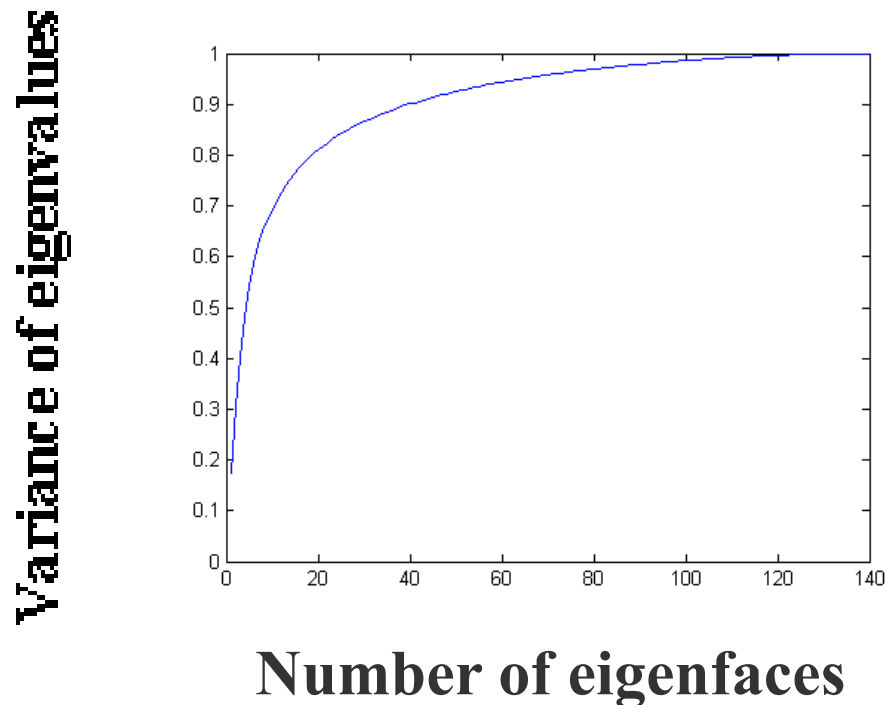
26 ASL Characters (S-Z)

Principal Component Analysis (5 of 10)

Example: PCA Formation



Example of eigenface images



Principal Component Analysis (6 of 10)

❖ Known images are projected on k-eigenimages to form weight vectors:

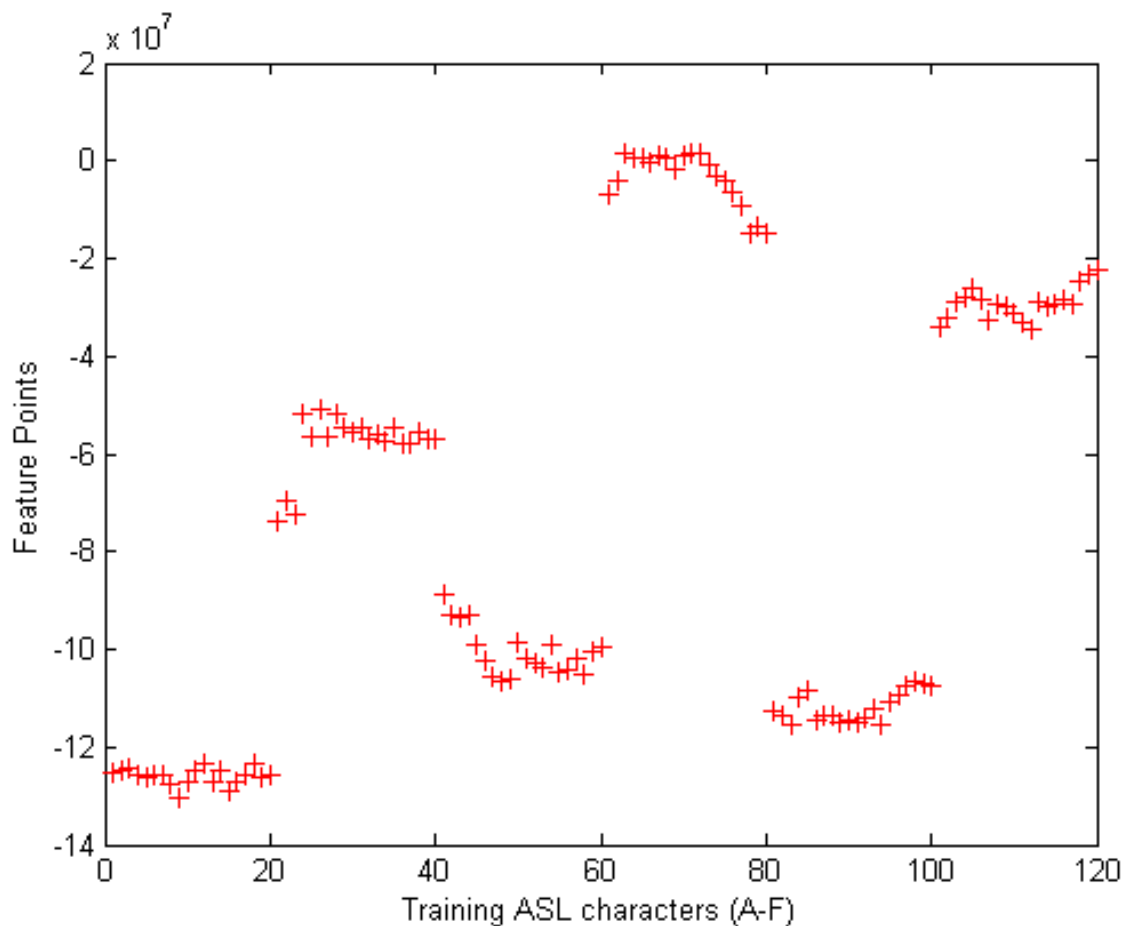
$$w_k = u_k^T (X_i - \Phi)$$

$$\Omega_i = [w_1, w_2, \dots, w_k]$$

Φ = Average image, $X_i = i^{th}$ Training image

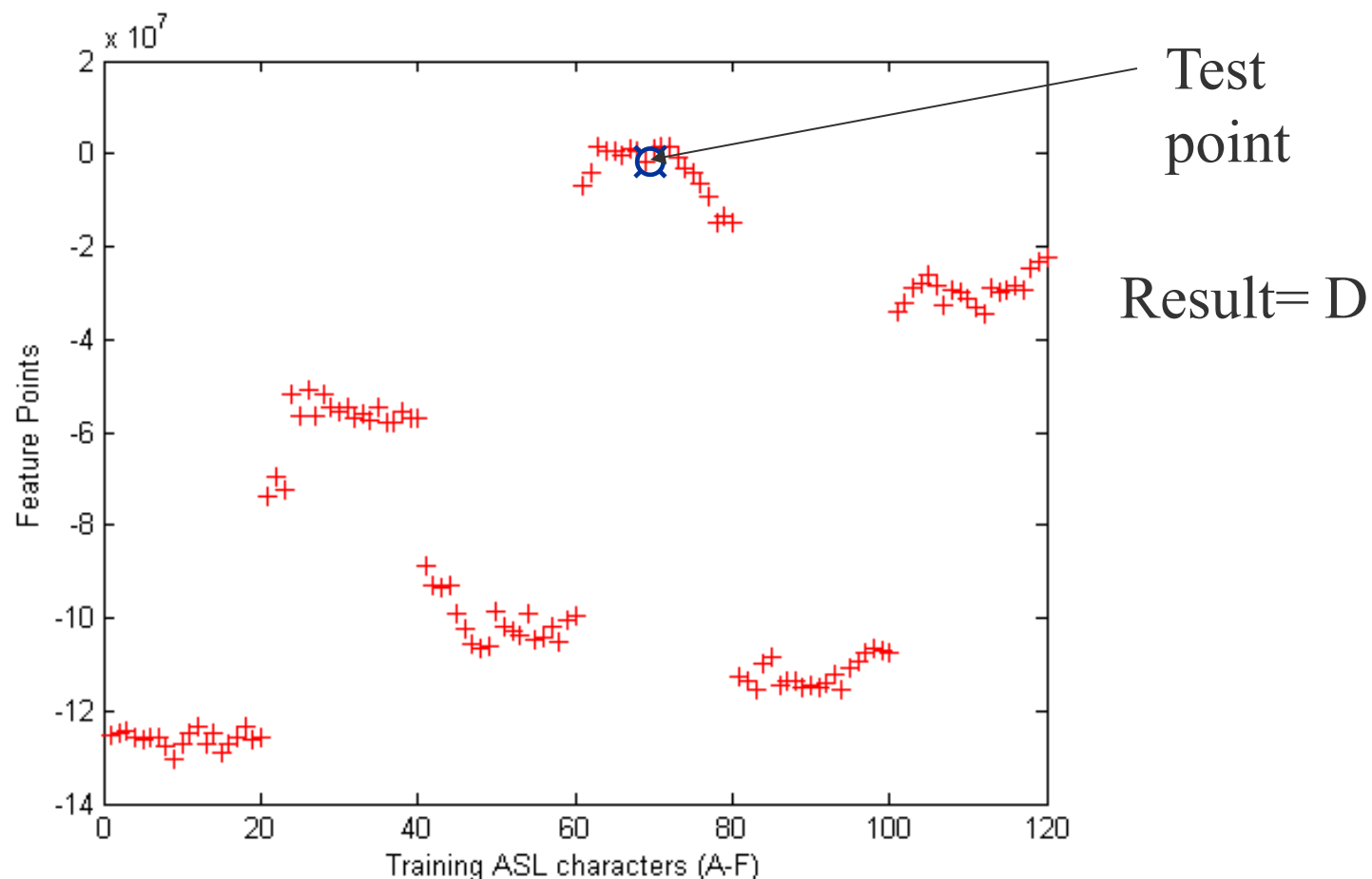
Principal Component Analysis (7 of 10)

Features Points for 6-ASL Characters



Principal Component Analysis (8 of 10)

Features Points for 6-ASL Characters



Principal Component Analysis (9 of 10)

- ❖ Each test image is projected on the k-eigenimages and calculated weight vectors (Ω) by
- ❖ Classify the pose based on minimum Euclidean distance.

$$\varepsilon_j = || \Omega - \Omega_i ||$$

$$\varepsilon = \arg \min \{ \varepsilon_j \}$$

Principal Component Analysis (10 of 10)

- ❖ If minimum Euclidian distance is lower than the predefined threshold value then corresponding pose is recognized.
 - i.e, ($\epsilon < Th_i$), then the image is classified as j^{th} pose

Subspace Method

- ❖ The main idea of the subspace method is similar to PCA
- ❖ In subspace method training images are grouped according to poses (classes) and eigenvectors for each class are calculated separately.
- ❖ In this method known images and test image are projected on each subspace separately.



Assignment: Group2

Next Class

- ❖ Introduction to Neural Network
- ❖ Architecture of a NN
- ❖ One application of NN to Recognize object



Questions? Or Suggestions?

Thanks to all